NOTES OF FUNDAMENTALS OF QUANTUM FIELD THEORY

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Chapter 1

Introduction

There are a few ways to quantize a theory. The familiar *canonical quantization method* consists in promoting the fields to operators, which satisfy some particular commutation relations. Observing then that fields are just an infinite sum of oscillators, we can then construct the Fock's space and treat interactions through perturbation theory. Feynman's diagrams are a fundamental tool to do visualize all the orders of interaction, from the tree-level up to whatever loop order we are interested in. However, the main problem is that loop contributions usually diverge, yielding to infinities that should be treated accordingly, mainly through the process of *renormalization*.

A different way to quantize a theory is the *Feynman quantization method*, which has the advantage that it is a really convenient way to derive correlation functions, the main object that we want to compute in Quantum Field Theory (QFT). In practice, we first need to define a *partition function*, or equivalently the *free-energy*, from where, using only functional derivatives, we compute the correlation functions that we need.

1.1 Reminiscences of Quantum Mechanics

In Quantum Mechanics (QM), an initial state $|q_i, t_i\rangle$ can evolve into a final state $|q_f, t_f\rangle$ through the time evolution operator

$$U(t_f - t_i) := e^{-iH(t_f - t_i)}. \tag{1.1}$$

In particular, we are interested in the probability that the initial state becomes the final state, in other words in

$$\langle q_f, t_f | U(t_f - t_i) | q_i, t_i \rangle \equiv \langle q_f, t_f | q_i, t_i \rangle. \tag{1.2}$$

To compute such an object, we first recall that

$$U(t_f - t_i) = U(t_f - t_n)U(t_n - t_{n-1}) \cdots U(t_2 - t_1)U(t_1 - t_i). \tag{1.3}$$

Using this property of the time evolution operator in (1.2) and the completion relation

$$\mathbb{I} = \int dq \; |q\rangle \langle q| \,, \tag{1.4}$$

we can write (1.2) as

$$\begin{aligned}\langle q_f, t_f|U(t_f - t_n)\mathbb{I}\cdots\mathbb{I}U(t_1 - t_i)|q_i, t_i\rangle &= \\ = \int dq_n \cdots \int dq_1 \langle q_f, t_f|U(t_f - t_n)|q_n, t_n\rangle \cdots \langle q_1, t_1|U(t_1 - t_i)|q_i, t_i\rangle.\end{aligned} \tag{1.5}$$

For simplicity, we define the time interval $\tau := (t_f - t_i)/n$, such that

$$U(\tau) = e^{-iH\tau}, \tag{1.6}$$

where the Hamiltonian H is given by

$$H = T(p) + V(q), \quad T(p) \equiv \frac{p^2}{2m}, \tag{1.7}$$

with the operators q and p satisfying the commutation relation $[q, p]$. Now the time evolution operator can then be rewritten as

$$U(\tau) = e^{-i[T(p)+V(q)]\tau} = e^{-iT(p)\tau}e^{-iV(q)\tau}e^{\pm i[T(p),V(q)]}\cdots, \tag{1.8}$$

where in the last step we used the BCH theorem. However, we can observe that the terms $e^{-iT(p)\tau}$ and $e^{-iV(q)\tau}$ are of order τ while the following one is of order τ^2, with all the following terms being of order even greater. Thus, in the limit where $\tau \to 0$, or equivalently for $n \to \infty$, the terms of order greater than τ are negligible. Hence,

$$U(\tau) \simeq e^{-iT(p)\tau}e^{-iV(q)\tau}. \tag{1.9}$$

Moreover, in this limit, let's compute the general term

$$\langle q_{j+1}, t_{j+1}|U(\tau)|q_j, t_j\rangle \simeq \langle q_{j+1}, t_{j+1}|e^{-iT(p)\tau}e^{-iV(q)\tau}|q_j, t_j\rangle. \tag{1.10}$$

Since, the states $|q_j, t_j\rangle$ are position eigenstates of the potential, we can use $e^{-iV(q)\tau}\,|q_j, t_j\rangle = e^{-iV(q_j)}\,|q_j, t_j\rangle$. Thus, we remain with

$$\langle q_{j+1}, t_{j+1}|U(\tau)|q_j, t_j\rangle \simeq \langle q_{j+1}, t_{j+1}|e^{-iT(p)\tau}|q_j, t_j\rangle e^{-iV(q_j)\tau}, \tag{1.11}$$

but now the states $|q_j, t_j\rangle$ are not eigenstates of the kinetic energy, so we can not use the same trick. On the other hand, we can instead use the previous trick of inserting a completion relation into the bracket, namely

$$\mathbb{I} = \int dp\ |p\rangle\,\langle p|\,, \tag{1.12}$$

to get

$$\begin{aligned}\langle q_{j+1}, t_{j+1}|e^{-iT(p)\tau}|q_j, t_j\rangle e^{-iV(q_j)\tau} &= \\ = \int dp_j \langle q_{j+1}, t_{j+1}|e^{ip^2/2m}|p_j\rangle\langle p_j|q_j, t_j\rangle e^{-iV(q_j)}.\end{aligned} \tag{1.13}$$

Now we can use the property of the eigenstates, so for a generic term we obtain

$$\langle q_{j+1}, t_{j+1}|U(\tau)|q_j, t_j\rangle = \int dp_j\, e^{-i[p^2/2m+V(q)]\tau} e^{2ip_j(q_{j+1}-q_j)}. \tag{1.14}$$

Following this process for every term results in the relation

$$\begin{aligned}&\langle q_f, t_f|U(t_f-t_i)|q_i, t_i\rangle \simeq \\ &\simeq \prod_{j=1}^{n}\int dq_j \int dp_j\, \exp\left\{-i\sum_{n=0}^{\infty}\left[\frac{p_j^2}{2m}+V(q_j)\right]\tau + ip_j(q_{j+1}-q_j)\right\}\end{aligned} \tag{1.15}$$

Moreover, multiplying the last term of the exponent with τ/τ and using the definition $\dot{q} = (q_{j+1}-q_j)/\tau$, we obtain

$$\begin{aligned}&\langle q_f, t_f|U(t_f-t_i)|q_i, t_i\rangle \simeq \\ &\quad\simeq \int\limits_{\substack{q(t_f)=q_f\\ q(t_i)=q_i}} \left(\prod_{j=1}^{n} dq_j dp_j\right)\exp\left\{-\int_{t_i}^{t_f} dt\left[\frac{p^2}{2m}+V(q)\right]-p\dot{q}\right\}.\end{aligned} \tag{1.16}$$

Noting that on dp we have a Gaussian integral and completing the square, at the end of the day, we arrive at

$$\begin{aligned}\langle q_f, t_f|U(t_f-t_i)|q_i, t_i\rangle &= \int\limits_{\substack{q(t_f)=q_f\\ q(t_i)=q_i}} \mathcal{D}q\, \exp\left\{i\int_{t_i}^{t_f} dt\left[\frac{1}{2}m^2\dot{q}^2 - V(q)\right]\right\} \\ &= \int \mathcal{D}q\, e^{iS/\hbar},\end{aligned} \tag{1.17}$$

where $\mathcal{D}q$ is the measure of the *path integral*, and where identified the action

$$S = \int_{t_i}^{t_f} dt\left[\frac{1}{2}m^2\dot{q}^2 - V(q)\right]. \tag{1.18}$$

In QM, the trajectory is not a well-defined concept: the particle simultaneously follows all the possible trajectories each with weight $e^{iS/\hbar}$. To retrieve Classical Mechanics, we need to send $\hbar \to 0$. In fact, more generally we have

$$\int dx\, e^{if(x)/\Lambda} \simeq e^{if(x_0)/\Lambda}\int dx\, e^{[if''(x_0)/\Lambda](x-x_0)^2}, \tag{1.19}$$

with x_0 a point such that $f'(x_0) = 0$. The dominant term then, for $\Lambda \to 0$, is $e^{if(x_0)/\Lambda}$. Hence, the classical solution corresponds to the one that extremises the action S, and it is the one that one gets from the dominant term when $\hbar \to 0$.

Suppose now that we want to compute the probability

$$\langle q_f, t_f|\mathcal{T}\{q(t_a)q(t_b)\}|q_i, t_i\rangle, \tag{1.20}$$

where, without loss of generality, we can assume that $t_a > t_b$, so that $\mathcal{T}\{q(t_a)q(t_b)\} = q(t_a)q(t_b)$. By following the same process previously shown, we will arrive at

$$\int \mathcal{D}q\, e^{iS/\hbar} q(t_a)q(t_b), \tag{1.21}$$

which holds even if $t_b > t_a$. Hence, more generally, we have

$$\int\limits_{\substack{q(t_f)=q_f \\ q(t_i)=q_i}} \mathcal{D}q\, e^{iS} q(t_1)\cdots q(t_n) = \langle q_f, t_f | \mathcal{T}\{q(t_1)\cdots q(t_n)\} | q_i, t_i \rangle. \tag{1.22}$$

1.2 Gaussian momenta generating functional

Now we discuss how to compute an important quantity that appears in QFT. The nth momentum of the Gaussian is defined as the integral

$$\int_{\mathbb{R}} dx\, e^{-Ax^2} x^n. \tag{1.23}$$

Clearly, when n is odd the integral is zero, instead when n is even the result is non-trivial. To compute such integrals, we usually make the change of variables $Ax^2 \equiv t$, and we use the Γ function. However, there is another way. Consider the integral

$$I(t) = \int_{\mathbb{R}} dx\, e^{-Ax^2 - xt} \tag{1.24}$$

which is just a "deformed" Gaussian integral with a *source term* xt. The main thing to observe is that $I(t)$ can be used to compute the momenta of the Gaussian through its derivatives, for example the first momenta is given by

$$\int_{\mathbb{R}} dx\, e^{-Ax^2} x = -\frac{d}{dt} I(t)\bigg|_{t=0}. \tag{1.25}$$

In general, the nth momenta in (1.23) can be computed as

$$\int_{\mathbb{R}} dx\, e^{-Ax^2} x^n = \bigg(-\frac{d}{dt} \bigg)^n I(t)\bigg|_{t=0}. \tag{1.26}$$

Therefore, we define the generating functional of all the momenta of the Gaussian as

$$I(t) = \frac{\int_{\mathbb{R}} dx\, e^{-Ax^2 - xt}}{\int_{\mathbb{R}} dx\, e^{-Ax^2}}, \tag{1.27}$$

where we have added a normalization such that $I(t = 0) = 1$. When we expand in Taylor series, we have

$$I(t) = \sum_{n=0}^{\infty} M_n(A) \frac{(-t)^n}{n!}, \tag{1.28}$$

where the coefficients $M_n(A)$, by definition, are

$$M_n(A) = \Big(-\frac{d}{dt} \Big)^n I(t)\Big|_{t=0}. \tag{1.29}$$

Consider now the exponent in (1.27) and complete the square using

$$\begin{aligned} I(t) &= \int_{\mathbb{R}} dx\, e^{-Ax^2 - xt} \\ &= \int_{\mathbb{R}} dx\, \exp\Big\{ -A\Big[x^2 + \frac{tx}{A} + \Big(\frac{t}{2A}\Big)^2\Big] + A\Big(\frac{t}{2A}\Big)^2 \Big\} \\ &= \int_{\mathbb{R}} dx\, \exp\Big\{ -A\Big(x + \frac{t}{2A}\Big)^2 + A\Big(\frac{t}{2A}\Big)^2 \Big\}, \end{aligned} \tag{1.30}$$

In this way, we get

$$I(t) = \frac{1}{\mathcal{N}} \int_{\mathbb{R}} dx\, e^{-A(a+t/4A)^2 + t^2/4A} = e^{t^2/4A}, \quad \mathcal{N} \equiv \int_{\mathbb{R}} dx\, e^{-Ax^2}, \tag{1.31}$$

where we have denoted the normalization as $\mathcal{N}$. With this simple expression for the generating functional we can compute some momenta; for example, the first one is

$$M_1 = -\frac{d}{dt} I(t)\Big|_{t=0} = -\frac{t}{2A} e^{t^2/4A}\Big|_{t=0} = 0, \tag{1.32}$$

as expected for all the odd momenta, while the second one is

$$M_2 = \frac{d^2}{dt^2} I(t)\Big|_{t=0} = \Big(\frac{1}{2A} + \frac{t^2}{4A^2}\Big) e^{t^2/4A}\Big|_{t=0} = -\frac{1}{2A}. \tag{1.33}$$

Obviously, one can use the previous momenta to compute the following one, for example

$$M_4 = \frac{d^4}{dt^4} I(t)\Big|_{t=0} = \frac{d^2}{dt^2} M_2\Big|_{t=0} = \frac{3}{4A^2}. \tag{1.34}$$

Now suppose that the generating functional is instead defined as

$$I(t) = \frac{1}{\mathcal{N}} \int_{\mathbb{R}} dx\, e^{-Ax^2 + f(x)}, \tag{1.35}$$

where now in place of xt we have a general function $f(x)$. Now through Taylor expansion we see that

$$e^{f(x)} = \sum_{n=0}^{\infty} c_n x^n, \tag{1.36}$$

since, at the end of the day, $e^{f(x)}$ can be seen as a function $g(x)$ which can be expanded with Taylor. Thus, using this expansion in the expression of $I(t)$, we have

$$I(t) = \frac{1}{\mathcal{N}} \sum_{n=0}^{\infty} c_n \int_{\mathbb{R}} dx\, e^{-Ax^2} x^n. \tag{1.37}$$

However, we know that

$$\int_{\mathbb{R}} e^{-Ax^2} x^n = \Big(- \frac{d}{dt} \Big)^n \int_{\mathbb{R}} dx\, e^{-Ax^2 - xt} \Big|_{t=0}. \tag{1.38}$$

Hence, we can rewrite $I(t)$ like

$$\begin{aligned} I(t) &= \sum_{n=0}^{\infty} c_n \Big(- \frac{d}{dt} \Big)^n \int_{\mathbb{R}} dx\, e^{-Ax^2 - xt} \Big|_{t=0} \\ &= \exp\left\{ f\Big(- \frac{d}{dt} \Big) \right\} I(t) \Big|_{t=0}, \end{aligned} \tag{1.39}$$

where in the last step we again used the Taylor series of the exponential function. All this discussion becomes quite useful if the generating functional is defined as

$$I(t) = \frac{1}{\mathcal{N}} \int_{\mathbb{R}} dx\, e^{-Ax^2 + \alpha f(x)}, \tag{1.40}$$

where $\alpha \in \mathbb{R}$. In particular, if $\alpha \ll 1$, we can use perturbation theory to write

$$\begin{aligned} I(t) &= \exp\left\{ \alpha f\Big(- \frac{d}{dt} \Big) \right\} I(t) \Big|_{t=0} \\ &= I(t)|_{t=0} + \alpha f\Big(- \frac{d}{dt} \Big) I(t) \Big|_{t=0} + \frac{\alpha^2}{2!} f\Big(- \frac{d}{dt} \Big)^2 I(t) \Big|_{t=0} + \cdots . \end{aligned} \tag{1.41}$$

1.3 Partition function

Previously, we defined in Quantum Mechanic a new quantization scheme involving the partition function

$$\mathcal{Z} = \int \mathcal{D}q\, e^{iS}, \quad S = \int_{\mathbb{R}} dt \left[\frac{1}{2} m\dot{q}^2 - V(q) \right]. \tag{1.42}$$

We now generalize to QFT by "promoting" the q operators to fields $\phi(x)$. Thus, the action becomes

$$S = \int d^4x\, \frac{1}{2} [\partial_\mu \phi(x) \partial^\mu \phi(x) - m^2 \phi(x)^2], \tag{1.43}$$

consequently, the partition function is

$$\mathcal{Z} = \int \mathcal{D}\phi\, e^{iS}. \tag{1.44}$$

Now, as we do in QFT, we want to compute the correlation function

$$\langle 0 | \mathcal{T}\{\phi(x_1) \cdots \phi(x_n)\} | 0 \rangle = \int \mathcal{D}\phi\, e^{iS} \phi(x_1) \cdots \phi(x_n). \tag{1.45}$$

For this purpose, we define the partition function

$$\mathcal{Z}[J] = \int \mathcal{D}\phi\, e^{iS + i\int d^4x\, J(x)\phi(x)}, \tag{1.46}$$

with an added source term dependent of $J(x)\phi(x)$. This definition obviously reduce to (1.44) by setting $J(x) = 0$. Consequently, the action get modified in a similar way

$$S = \int d^4x\, \frac{1}{2}[\partial_\mu\phi(x)\partial^\mu\phi(x) - m^2\phi(x)^2 + J(x)\phi(x)]. \tag{1.47}$$

To derive the equations of motion (EOMs), we need to compute the Euler-Lagrange equations

$$\partial_\mu\left(\frac{\delta\mathcal{L}}{\delta(\partial_\mu\phi)}\right) - \frac{\delta\mathcal{L}}{\delta\phi} = 0, \tag{1.48}$$

where the Lagrangian is of course the integrand of the previous action, namely

$$\mathcal{L} = \frac{1}{2}[\partial_\mu\phi(x)\partial^\mu\phi(x) - m^2\phi(x)^2 + J(x)\phi(x)]. \tag{1.49}$$

The computation results in

$$(\Box + m^2)\phi(x) = J(x), \quad \Box \equiv \partial_\mu\partial^\mu. \tag{1.50}$$

This is precisely the Klein-Gordon equation when we set $J(x) = 0$, also for this reason it is clear we $J(x)$ is called a source term.

Notation 1. *To make the notation a bit lighter, we can use* $\langle J\phi\rangle_x = \int d^4x\, J(x)\phi(x)$. *More generally, we set*

$$\langle(\cdots)\rangle_{xy\cdots z} \equiv \left(\int d^4x \int d^4y \cdots \int d^4z\right)(\cdots). \tag{1.51}$$

For the ordinary derivative ∂_i we have the property $\partial_i x^j = \delta_i^j$. Similarly, for the functional derivative holds the property

$$\frac{\delta\phi(y)}{\delta\phi(x)} = \delta^4(x - y). \tag{1.52}$$

Knowing this, we can see that

$$\begin{aligned} -i\frac{\delta}{\delta J(x)}\mathcal{Z}[J] &= \int \mathcal{D}\phi \left(-i\frac{\delta}{\delta J(y)}e^{i\langle J\phi\rangle_x}\right)e^{iS} \\ &= \int \mathcal{D}\phi\, \phi(y) e^{i(S + \langle J\phi\rangle_x)}. \end{aligned} \tag{1.53}$$

Setting $J(x) = 0$ at the end of the computation returns exactly what we originally wanted to compute. Therefore, we can easily see that

$$\langle 0|\mathcal{T}\{\phi(x_1)\cdots\phi(x_n)\}|0\rangle = \prod_{i=1}^{n}\left(-i\frac{\delta}{\delta J(x_i)}\right)\mathcal{Z}[J]\Big|_{J=0}. \tag{1.54}$$

The partition function $\mathcal{Z}[J]$ can then be seen as the generating functional of the Green functions, namely

$$\mathcal{Z}[J] = \sum_{n=0}^{\infty}\frac{i^n}{n!}\langle G_n(x_1,\ldots,x_n)J(x_1)\cdots J(x_n)\rangle_{x_1\cdots x_n}. \tag{1.55}$$

1.3.1 Free partition function

All of this is very useful if we can compute the partition function. Let's try to do it for the free field. For this purpose we indicate the partition function as $\mathcal{Z}_0[J]$. Focusing on the exponent, we have

$$S_0 + \langle J\phi \rangle_x = \int d^4x \left(\frac{1}{2}\partial_\mu \phi \partial^\mu \phi - \frac{1}{2}m^2\phi^2 + J\phi \right). \tag{1.56}$$

We now want to complete the square as we did before. However, to do it we first need to write the field $\phi(x)$ through its Fourier transform

$$\phi(x) = \int \frac{d^4k}{(2\pi)^4}\, e^{ikx} \tilde{\phi}(k), \tag{1.57}$$

and also write the source term in an analogous way

$$J(x) = \int \frac{d^4k}{(2\pi)^4}\, e^{ikx} \tilde{J}(k). \tag{1.58}$$

Substituting these in the exponent gets us

$$\begin{aligned} S_0 + \langle J\phi \rangle_x = \int d^4x \int \frac{d^4k}{(2\pi)^4} \int \frac{d^4\tilde{k}}{(2\pi)^4} \Big\{ &- k\tilde{k}\tilde{\phi}(k)\tilde{\phi}(\tilde{k}) - \frac{m^2}{2}\tilde{\phi}(k)\tilde{\phi}(\tilde{k}) \\ &+ \frac{1}{2}[\tilde{J}(k)\tilde{\phi}(\tilde{k}) + \tilde{J}(\tilde{k})\tilde{\phi}(k)] \Big\} e^{i(k+\tilde{k})x}. \end{aligned} \tag{1.59}$$

In the latter relation, we identify the definition of the Dirac delta, namely

$$\int \frac{d^4x}{(2\pi)^4} e^{i(k+\tilde{k})x} = \delta(k - \tilde{k}). \tag{1.60}$$

With it, the said expression simplifies to

$$S_0 + \langle J\phi \rangle_x = \frac{1}{2} \int \frac{d^4k}{(2\pi)^4} \Big[(k^2 - m^2)\tilde{\phi}(-k)\tilde{\phi}(k) + \tilde{J}(-k)\tilde{\phi}(k) + \tilde{J}(k)\tilde{\phi}(-k) \Big]. \tag{1.61}$$

Now we can complete the square

$$\begin{aligned} S_0 + \langle J\phi \rangle_x = \frac{1}{2} \int \frac{d^4k}{(2\pi)^4} \Big\{ &[\tilde{\phi}(-k) + \tilde{J}(-k)(k^2 - m^2)^{-1}](k^2 - m^2) \\ &\times [\tilde{\phi}(k) + (k^2 - m^2)^{-1}\tilde{J}(k)] - \tilde{J}(-k)(k^2 - m^2)^{-1}\tilde{J}(k) \Big\}. \end{aligned} \tag{1.62}$$

To simplify this expression, we can shift the field using

$$\tilde{\phi} + (k^2 - m^2)^{-1}\tilde{J}(x) \mapsto \tilde{\phi}, \tag{1.63}$$

thus getting

$$S_0 + \langle J\phi\rangle_x = \frac{1}{2}\int \frac{d^4k}{(2\pi)^4}\,\{\tilde{\phi}(-k)(k^2-m^2)\tilde{\phi}(k) - \tilde{J}(-k)(k^2-m^2)^{-1}\tilde{J}(k)\}, \tag{1.64}$$

where since the translation (1.63) has a unitary Jacobian, the integral's measure does not change. Therefore,

$$\begin{aligned}\mathcal{Z}_0[J] = \int \mathcal{D}\phi\, \exp\left\{\frac{i}{2}\int \frac{d^4k}{(2\pi)^4}\tilde{\phi}(-k)(k^2-m^2)\tilde{\phi}(k)\right\} \\ \times \exp\left\{-\frac{i}{2}\int \frac{d^4k}{(2\pi)^4}\tilde{J}(-k)(k^2-m^2)^{-1}\tilde{J}(k)\right\}.\end{aligned} \tag{1.65}$$

Going back to coordinate space from momentum space through the anti-Fourier transforms

$$\tilde{\phi}(k) = \int d^4x\, e^{-ikx}\phi(x), \quad J(x) = \int d^4x\, e^{-ikx}J(x), \tag{1.66}$$

we get the exponent

$$\begin{aligned}&\int \frac{d^4k}{(2\pi)^4}\int d^4x \int d^4y\,(k^2-m^2)\phi(x)\phi(y)e^{-ik(x-y)} = \\ &= \int \frac{d^4k}{(2\pi)^4}\int d^4x \int d^4y\,[\partial_\mu\phi(x)\partial^\mu\phi(y) - m^2\phi(x)\phi(y)]e^{ik(x-y)},\end{aligned} \tag{1.67}$$

where we have used the fact that

$$k^2 e^{ik(x-y)} = \frac{\partial}{\partial x}\frac{\partial}{\partial y}e^{-ik(x-y)}. \tag{1.68}$$

Then, using again the definition of the Dirac delta (1.60), we obtain

$$\begin{aligned}&\int \frac{d^4k}{(2\pi)^4}\int d^4x \int d^4y\,(k^2-m^2)\phi(x)\phi(y)e^{-ik(x-y)} = \\ &\qquad = \int d^4x\,[\partial_\mu\phi\partial^\mu\phi - m^2\phi(x)^2].\end{aligned} \tag{1.69}$$

Therefore, $\mathcal{Z}_0[J]$ becomes

$$\mathcal{Z}_0[J] = \int \mathcal{D}\phi\, e^{iS_0}\exp\left\{-\frac{i}{2}\int \frac{d^4k}{(2\pi)^4}\tilde{J}(-k)(k^2-m^2)^{-1}\tilde{J}(k)\right\}. \tag{1.70}$$

Including the normalization $\mathcal{N} \equiv \int \mathcal{D}\phi\, e^{iS_0}$ such that $\mathcal{Z}_0[J=0] = 1$, we obtain

$$\begin{aligned}\mathcal{Z}_0[J] &= \frac{1}{\mathcal{N}}\int \mathcal{D}\phi\, e^{i(S_0+\langle J\phi\rangle_x)} \\ &= \exp\left\{-\frac{i}{2}\int \frac{d^4k}{(2\pi)^4}\,\tilde{J}(-k)(k^2-m^2)^{-1}\tilde{J}(k)\right\} \\ &= \exp\left\{-\frac{1}{2}\int d^4x\int d^4y\,J(x)\left(\int \frac{d^4k}{(2\pi)^4}\,\frac{ie^{-ik(x-y)}}{k^2-m^2}\right)J(y)\right\},\end{aligned} \tag{1.71}$$

where the term in the round parentheses can be identified to be the *Feynman propagator*

$$\Delta_{\mathrm{F}}(x-y) = \int \frac{d^4k}{(2\pi)^4}\, \frac{ie^{-ik(x-y)}}{k^2-m^2}. \tag{1.72}$$

Notation 2. *More often than not, the Feynman propagator will be denoted as* $\Delta_{\mathrm{F}}(x-y) \equiv \Delta(x-y)$*, dropping the* F *subscript.*

Using this latter relation and the shorthand notation for the integrals, we obtain

$$\mathcal{Z}_0[J] = \exp\left\{ -\frac{1}{2}\langle J(x)\Delta(x-y)J(y)\rangle_{xy} \right\}. \tag{1.73}$$

To be a bit more precise, the integral that defines the free partition function is not really a Gaussian integral since the exponent does not behave properly, especially when sending $J(x) = 0$ it oscillates really quickly, making the integral to not converge. Therefore, to regain some sort of convergence, we add a convergence term $i\epsilon$ in the exponent, namely

$$\mathcal{Z}_0[J] = \frac{1}{\mathcal{N}} \int \mathcal{D}\phi\, \exp\left\{ i \int d^4x \left(\frac{1}{2}\partial_\mu\phi\partial^\mu\phi - \frac{1}{2}(m^2 - i\epsilon)\phi^2 + J\phi \right) \right\}, \tag{1.74}$$

also the Feynman propagator's definition becomes

$$\Delta_{\mathrm{F}}(x-y) = \int \frac{d^4k}{(2\pi)^4}\, \frac{ie^{-ik(x-y)}}{k^2-m^2+i\epsilon}. \tag{1.75}$$

However, the proper way to do this is by a *Wick rotation*, that is setting $\tau = \pm it$, where the sign of this rotation is yet to be decided. Either way, the idea consist in the fact that a Wick rotation implies sending $\eta_{\mu\nu} \mapsto -\delta_{\mu\nu}$, in other words the metric becomes Euclidean, so

$$ds^2 = dt^2 - d\boldsymbol{x}^2 = -(dt^2 + d\boldsymbol{x}^2), \tag{1.76}$$

where the overall minus sign appears because of the choice for the Minkowski metric $\eta_{\mu\nu} = \mathrm{diag}(+1,-1,-1,-1)$, using the other convention would have yielded to the standard Euclidean metric. In this way, we have

$$\frac{i}{2}\langle \partial_\mu\phi\partial^\mu\phi - m^2\phi^2\rangle_x \overset{\tau=\pm it}{\longmapsto} -i(\mp i)\int d^4x_{\mathrm{E}}\, \frac{1}{2}(\partial_\mu\phi\partial^\mu\phi - m^2\phi^2). \tag{1.77}$$

Therefore, the Euclidean free partition function can be written as

$$\mathcal{Z}_0^{\mathrm{E}}[J] = \int \mathcal{D}\phi\, \exp\{-(S_{\mathrm{E}} + \langle J\phi\rangle_{x_{\mathrm{E}}})\}. \tag{1.78}$$

Before continuing to the interacting theory, we just summarize the main result obtained for the free partition function and discuss some examples of

Green's functions. So far we have found that

$$\begin{aligned}\mathcal{Z}_0[J] &= \frac{\int \mathcal{D}\phi\, e^{i(S_0+\langle J\phi\rangle_x)}}{\int \mathcal{D}\phi\, e^{iS_0}} \\ &= \exp\left\{ -\frac{1}{2}\langle J(x)\Delta(x-y)J(y)\rangle_{xy}\right\} \\ &= \sum_{n=0}^{\infty}\frac{i^n}{n!}\langle G_n(x_1,\dots,x_n)J(x_1)\cdots J(x_n)\rangle_{x_1\cdots x_n},\end{aligned} \tag{1.79}$$

where

$$G_n(x_1,\dots,x_n) = (-i)^n\left(\frac{\delta^n}{\delta J(x_1)\cdots\delta J(x_n)}\right)\mathcal{Z}_0[J]\Big|_{J=0}. \tag{1.80}$$

Notation 3. *In the following, we shall often use $\phi(x_i)\equiv\phi_i$, $J(x_i)\equiv J_i$, $\Delta_{\rm F}(x-y)\equiv\Delta_{xy}$, and $\Delta_{\rm F}(x_i-x_j)\equiv\Delta_{ij}$ to make the notation a bit lighter.*

Example 1 (some Green's functions)**.** *We compute the first two Green's functions pretty easily. The first one is*

$$\begin{aligned}G_1(x_1) = -i\frac{\delta}{\delta J_1}\mathcal{Z}_0[J]\Big|_{J=0} &= (-i)\left(-\frac{1}{2}\right)\frac{\delta}{\delta J_1}(\langle J(x)\Delta_{xy}J(y)\rangle_{xy})\mathcal{Z}_0[J]\Big|_{J=0} \\ &= i\left\langle\frac{\delta J(x)}{\delta J_1}\Delta_{xy}J(y)\right\rangle_{xy}\mathcal{Z}_0[J]\Big|_{J=0} \\ &= i\langle\Delta_{1y}J(y)\rangle_y\mathcal{Z}_0[J]\Big|_{J=0} \\ &= 0,\end{aligned} \tag{1.81}$$

as we expect for odd momenta. On the other side, the 2-point Green's function, using the result obtained for $G_1(x_1)$, is

$$\begin{aligned}G_2(x_1,x_2) &= -\frac{\delta^2}{\delta J_1\delta J_2}\mathcal{Z}_0[J]\Big|_{J=0} \\ &= -\frac{\delta}{\delta J_2}(\langle\Delta_{xy}J(y)\rangle_y\mathcal{Z}_0[J])\Big|_{J=0} \\ &= -\Delta_{12}\mathcal{Z}_0[J]|_{J=0} - \langle\Delta_{1y}J(y)\rangle_y\frac{\delta}{\delta J_2}\mathcal{Z}_0[J]\Big|_{J=0} \\ &= -\Delta_{12}\mathcal{Z}_0[J]|_{J=0} + \langle\Delta_{1y}J(y)\rangle_y\langle\Delta_{2w}J(w)\rangle_w\mathcal{Z}_0[J]|_{J=0} \\ &= -\Delta(x_1-x_2),\end{aligned} \tag{1.82}$$

so it corresponds to the Feynman propagator.

Another important Green's function, other than the ones in the example,

is the 4-point Green's function $G_4(x_1, x_2, x_3, x_4)$. This is given by

$$\begin{aligned}
G_4(x_1,x_2,x_3,x_4) &= \frac{\delta^4}{\delta J_1 \delta J_2 \delta J_3 \delta J_4} \mathcal{Z}_0[J]\Big|_{J=0} \\
&= \frac{\delta^3}{\delta J_1 \delta J_2 \delta J_3}(-\langle \Delta_{4y} J_y \rangle_y) \mathcal{Z}_0[J] \\
&= \frac{\delta^2}{\delta J_1 \delta J_2}(-\Delta_{43} + \langle \Delta_{4y} J_y \rangle_y \langle \Delta_{3w} J_w \rangle_w) \mathcal{Z}_0[J] \\
&= \frac{\delta}{\delta J_1}(\Delta_{43}\langle \Delta_{2y} J_y \rangle_y + \Delta_{42}\langle \Delta_{3w} J_w \rangle_w + \langle \Delta_{4y} J_y \rangle_y \Delta_{32} \\
&- \langle \Delta_{4y} J_y \rangle_y \langle \Delta_{3w} J_w \rangle_w \langle \Delta_{2z} J_z \rangle_z) \mathcal{Z}_0[J] \\
&= [\Delta_{43}\Delta_{21} - \Delta_{43}\langle \Delta_{2y} J_y \rangle_y \langle \Delta_{1w} J_w \rangle_w + \Delta_{42}\Delta_{31} \\
&- \Delta_{42}\langle \Delta_{3w} J_w \rangle_w \langle \Delta_{1y} J_y \rangle_y + \Delta_{32}\Delta_{41} \\
&- \Delta_{32}\langle \Delta_{4y} J_y \rangle_y \langle \Delta_{1w} J_w \rangle_w - \Delta_{41}\langle \Delta_{3w} J_w \rangle_w \langle \Delta_{2y} J_y \rangle_y \\
&- \langle \Delta_{4y} J_y \rangle_y \Delta_{31} \langle \Delta_{2w} J_w \rangle_w - \langle \Delta_{4y} J_y \rangle_y \langle \Delta_{3w} J_w \rangle_w \Delta_{21} \\
&+ \langle \Delta_{4y} J_y \rangle_y \langle \Delta_{3w} J_w \rangle_w \langle \Delta_{2z} J_z \rangle_z \langle \Delta_{1k} J_k \rangle_k]|_{J=0} \\
&= \Delta_{43}\Delta_{21} + \Delta_{42}\Delta_{31} + \Delta_{41}\Delta_{32},
\end{aligned} \tag{1.83}$$

where in the last step we set $J = 0$, which makes the majority of the terms to vanish.

Notation 4. *In the following, the condition $(\cdots)|_{J=0}$ is understood during the intermediate steps of the computations, so we explicitly write it usually at the begging and at the end of them.*

The latter result for the 4-point function is really important since it shows us that Wick's theorem is automatically included at the end of the computation. In fact, if we compute the time ordered product of four fields, we get

$$\begin{aligned}
\langle 0|\mathcal{T}\{\phi_1\phi_2\phi_3\phi_4\}|0\rangle &= \langle 0|\,(:\phi_1\phi_2\phi_3\phi_4: + :\overbracket{\phi_1\phi_2}\phi_3\phi_4: + :\overbracket{\phi_1\phi_2\phi_3}\phi_4: \\
&+ :\overbracket{\phi_1\phi_2\phi_3\phi_4}: + :\phi_1\overbracket{\phi_2\phi_3}\phi_4: + :\phi_1\overbracket{\phi_2\phi_3\phi_4}: \\
&+ :\phi_1\phi_2\overbracket{\phi_3\phi_4}: + :\overbracket{\phi_1}\phi_2\overbracket{\phi_3}\phi_4: + :\overbracket{\phi_1}\overbracket{\phi_2\phi_3}\overbracket{\phi_4}: \\
&+ :\overbracket{\phi_1\phi_2}\overbracket{\phi_3\phi_4}:)\,|0\rangle \\
&= \langle 0|:\overbracket{\phi_1}\overbracket{\phi_2}\overbracket{\phi_3}\phi_4: + :\overbracket{\phi_1\phi_2\phi_3}\phi_4: + :\overbracket{\phi_1\phi_2\phi_3\phi_4}:|0\rangle,
\end{aligned} \tag{1.84}$$

where $:(\cdots):$ is the *normal ordering operator*. Moreover, one can also represent the Green's functions in a pictorial way; for example, $G_2(x_1, x_2)$ and $G_4(x_1, x_2, x_3, x_4)$ can be expressed as the following diagrams

$$+FD. \tag{1.85}$$

By looking at those, we can see that all the diagrams, expect the one for the 2-point function, are *disconnected diagrams*[1]. This means that the free partition function is the generator of all the diagrams, both connected and disconnected. However, we are interested in connected diagrams for the most part, so we now ask ourselves if there is a way to get only the connected diagrams, while removing the disconnected ones. For this purpose, we define

$$\mathcal{Z}[J] = e^{-\mathcal{W}[J]}, \tag{1.86}$$

where the functional

$$\mathcal{W}[J] = \sum_{n=0}^{\infty} \frac{1}{n!} \langle G_n^{\mathrm{c}}(x_1, \dots, x_n) J_1 \cdots J_n \rangle_{x_1 \cdots x_n} \tag{1.87}$$

is the functional that generates all the connected diagrams. With this, it follows that

$$G^{\mathrm{c}}(x_1, \dots, x_n) = \frac{\delta^n}{\delta J_1 \cdots \delta J_n} \mathcal{W}[J]. \tag{1.88}$$

For the free theory, so using $\mathcal{Z}_0[J]$ and defining $\mathcal{W}_0[J]$, the exact result is

$$\mathcal{W}_0[J] = \frac{i}{2} \langle J_x \Delta_{xy} J_y \rangle_{xy}, \tag{1.89}$$

which describes only the propagation between x_1 and x_2. Thus, for the free theory, we have

$$G^{\mathrm{c}}(x_1, \dots, x_n) \equiv G_2^{\mathrm{c}}(x_1, x_2) = \Delta_{21}. \tag{1.90}$$

1.3.2 Interacting partition function

To include some sort of interaction in our theory, we need to introduce a potential $V(\phi)$ in the free action, namely we need to use

$$S = \int d^4x \left[\frac{1}{2} \partial_\mu \partial^\mu \phi - \frac{1}{2} m^2 \phi^2 + V(\phi) \right] = S_0 - \int d^4x \, V(\phi). \tag{1.91}$$

Then, the interacting partition function is

$$\mathcal{Z}[J] = \frac{\int \mathcal{D}\phi \, e^{i(S + \langle J\phi \rangle_x)}}{\int \mathcal{D}\phi \, e^{iS_0}} = \frac{\int \mathcal{D}\phi \, e^{-i\langle V(\phi) \rangle_x} e^{i(S_0 + \langle J\phi \rangle_x)}}{\int \mathcal{D}\phi \, e^{iS_0}}. \tag{1.92}$$

As we did before, we can expand in Taylor series the exponential of the potential as $e^{-i\langle V(\phi) \rangle_x} = \sum_{n=0}^{\infty} c_n \phi^n$, so that the partition function becomes

$$\begin{aligned} \mathcal{Z}[J] &= \frac{1}{\mathcal{N}} \sum_{n=0}^{\infty} c_n \int \mathcal{D}\phi \, \phi^n e^{i(S_0 + \langle J\phi \rangle_x)} \\ &= \frac{1}{\mathcal{N}} \sum_{n=0}^{\infty} c_n \int \mathcal{D}\phi \left(-i \frac{\delta}{\delta J} \right)^n e^{i(S_0 + \langle J\phi \rangle_x)}, \end{aligned} \tag{1.93}$$

[1] Here with "disconnected" we refer to all the diagrams that to be drawn require to lift the pen from the paper.

then, by commuting the path integral with the functional derivative and using the definition of the exponential's Taylor series, we obtain

$$\begin{aligned}\mathcal{Z}[J] &= \frac{1}{\mathcal{N}} \exp\left\{ \left\langle -iV\left(-i\frac{\delta}{\delta J}\right)\right\rangle_x \right\} \int \mathcal{D}\phi\, e^{i(S_0+\langle J\phi\rangle_x)} \\ &= \frac{1}{\mathcal{N}} \exp\left\{ \left\langle -iV\left(-i\frac{\delta}{\delta J}\right)\right\rangle_x \right\} \mathcal{Z}_0[J].\end{aligned} \tag{1.94}$$

In this way, the functional integral can be computed through perturbation theory if the potential is such that $V(\phi) := \lambda\hat{V}(\phi)$ with $\lambda \ll 1$. In particular, denoting for simplicity

$$\delta \equiv \exp\left\{ \left\langle -iV\left(-i\frac{\delta}{\delta J}\right)\right\rangle_x \right\}, \tag{1.95}$$

so that $\mathcal{Z}[J] = \delta\mathcal{Z}_0[J]$, we can write

$$\begin{aligned}\delta\mathcal{Z}_0[J] = \delta e^{-\mathcal{W}_0[J]} &= e^{-\mathcal{W}_0[J]} e^{\mathcal{W}_0[J]} \delta e^{-\mathcal{W}_0[J]} \\ &= e^{-\mathcal{W}_0[J]} e^{\mathcal{W}_0[J]} (1+\delta-1) e^{-\mathcal{W}_0[J]} \\ &= e^{-\mathcal{W}_0[J]} + e^{\mathcal{W}_0[J]} (\delta-1) e^{-\mathcal{W}_0[J]},\end{aligned} \tag{1.96}$$

therefore

$$\begin{aligned}-\ln \mathcal{Z}[J] = \mathcal{W}[J] &= \mathcal{W}_0[J] - \ln\{e^{\mathcal{W}_0[J]}[1+(\delta-1)]e^{-\mathcal{W}_0[J]}\} \\ &= -\ln[1 + e^{\mathcal{W}_0[J]}(\delta-1)e^{\mathcal{W}_0[J]}].\end{aligned} \tag{1.97}$$

Now denoting $A \equiv e^{\mathcal{W}_0[J]}(\delta-1)e^{-\mathcal{W}_0[J]}$, we can expand the logarithm in Taylor series

$$\ln(1+A) = A - \frac{A^2}{2} + \frac{A^3}{3!} + \cdots, \tag{1.98}$$

where A can itself be expanded as

$$A = e^{\mathcal{W}_0[J]}(\lambda\alpha_1 + \lambda^2\alpha_2 + \lambda^3\alpha_3 + \cdots)e^{-\mathcal{W}_0[J]}, \tag{1.99}$$

with

$$\alpha_i \equiv \left\langle -iV\left(-i\frac{\delta}{\delta J}\right)\right\rangle_x. \tag{1.100}$$

Putting all together, we have

$$\begin{aligned}\mathcal{W}[J] &= \mathcal{W}_0[J] - (\lambda\alpha_1 + \lambda^2\alpha_2 + \lambda^3\alpha_3 + \cdots) \\ &\quad + \frac{1}{2}(\lambda\alpha_1 + \lambda^2\alpha_2 + \lambda^3\alpha_3 + \cdots)^2 \\ &\quad + \frac{1}{3!}(\lambda\alpha_1 + \lambda^2\alpha_2 + \lambda^3\alpha_3 + \cdots)^3 \\ &\simeq \mathcal{W}_0[J] - \lambda\alpha_1 - \lambda^2\left(\alpha_2 - \frac{\alpha_1^2}{2}\right) - \lambda^3\left(\alpha_3 - \alpha_1\alpha_2 + \frac{\alpha_1^3}{3}\right),\end{aligned} \tag{1.101}$$

where in the last step we stopped at the third order in λ. To compute the α_is we need to assume, in a more precise way, the form of the potential. We

choose $V(\phi) \equiv \lambda\phi^4/4$ which defines the ϕ^4-*theory*, so that, for example, α_1 is

$$\alpha_1 = -\frac{i}{4!} e^{\mathcal{W}_0[J]} \left\langle \frac{\delta^4}{\delta J_x^4} \right\rangle_x e^{-\mathcal{W}_0[J]}. \tag{1.102}$$

Before compute it, we can see that the partition function in this particular theory is

$$\begin{aligned}
\mathcal{Z}[J] &= \sum_{n=0}^{\infty} \frac{(-i)^n}{n!} \left(\prod_{i=1}^{n} \int d^4x_i \right) \int \mathcal{D}\phi \, \phi_1^4 \cdots \phi_n^4 e^{i(S_0 + \langle J\phi \rangle_x)} \\
&= \sum_{n=0}^{\infty} \frac{(-i)^n}{n!} \left(\prod_{i=1}^{n} \int d^4x_i \right) \frac{\delta^4}{\delta J_1^4} \cdots \frac{\delta^4}{\delta J_n^4} \int \mathcal{D}\phi \, e^{i(S_0 + \langle J\phi \rangle_x)} \\
&= \sum_{n=0}^{\infty} \frac{(-i)^n}{n!} \left(\prod_{i=1}^{n} \int d^4x_i \frac{\delta^4}{\delta J_i^4} \right) \int \mathcal{D}\phi \, e^{i(S_0 + \langle J\phi \rangle_x)} \\
&= \exp\left\{ -i \left\langle \frac{\delta^4}{\delta J_x^4} \right\rangle_x \right\} \int \mathcal{D}\phi \, e^{i(S_0 + \langle J\phi \rangle_x)}
\end{aligned} \tag{1.103}$$